

A Glimpse of Computability Theory in Fractal Geometry

Zach Archibald

November 2024

Decidability

Decision Problem

A problem whose answer is yes or no

Decidability

Decision Problem

A problem whose answer is yes or no

Example

Given a list of real numbers N and a real number x , does N contain x ?

$$N = \begin{array}{|c|c|c|c|} \hline n_1 & n_2 & \dots & n_k \\ \hline \end{array} \quad \text{query: } x$$

Decidability

Decision Problem

A problem whose answer is yes or no

Example

Given a list of real numbers N and a real number x , does N contain x ?

$$N = \boxed{\begin{array}{|c|c|c|c|} \hline n_1 & n_2 & \cdots & n_k \\ \hline \end{array}} \quad \text{query: } x$$

Decidable Problem

A decision problem is considered decidable if and only if there exists an algorithm that solves any instance of the problem in finite time.

Decidability

Decision Problem

A problem whose answer is yes or no

Example

Given a list of real numbers N and a real number x , does N contain x ?

$$N = \boxed{\begin{array}{|c|c|c|c|} \hline n_1 & n_2 & \cdots & n_k \\ \hline \end{array}} \quad \text{query: } x$$

Decidable Problem

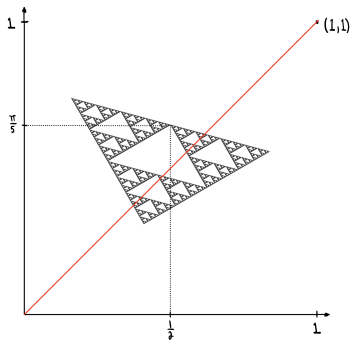
A decision problem is considered decidable if and only if there exists an algorithm that solves any instance of the problem in finite time.

Example

The above example is decidable by the following algorithm: For any N and x , check each index of N for equality with x . If equality is found, return yes. Otherwise, return no.

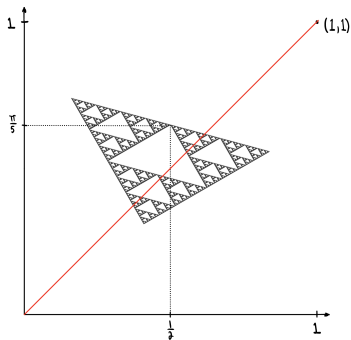
A Decision Problem in Fractal Geometry

Given an iterated function system acting on $[0, 1]^2$, does its attractor $A_{\mathcal{W}}$ intersect the line segment between $\langle 0, 0 \rangle$ and $\langle 1, 1 \rangle$?



A Decision Problem in Fractal Geometry

Given an iterated function system acting on $[0, 1]^2$, does its attractor $A_{\mathcal{W}}$ intersect the line segment between $\langle 0, 0 \rangle$ and $\langle 1, 1 \rangle$?



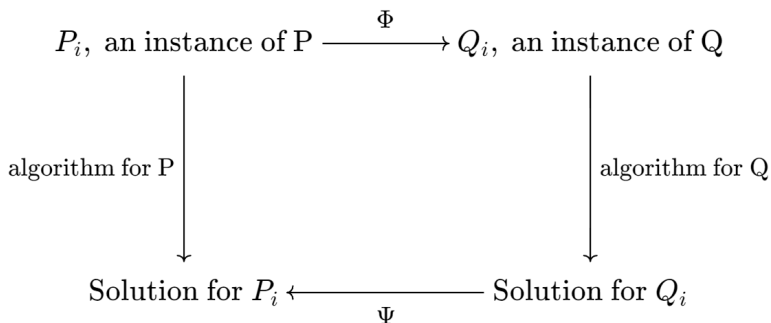
We will determine whether or not this problem is decidable.

Reductions

We say a problem P reduces to problem Q if there exist algorithms Φ and Ψ which satisfy the following diagram:

Reductions

We say a problem P reduces to problem Q if there exist algorithms Φ and Ψ which satisfy the following diagram:



Reduction Example

Let N be as before. The problem of identifying the maximal element of N reduces to the problem of sorting:

Reduction Example

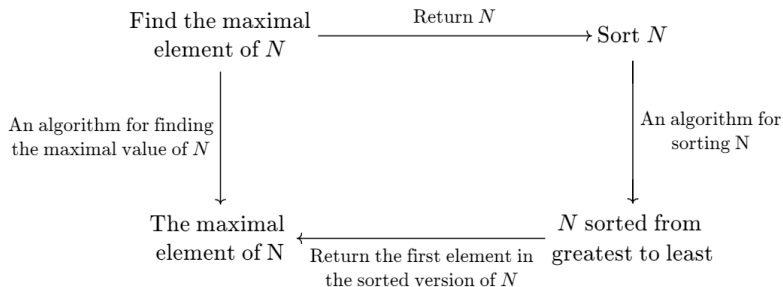
Let N be as before. The problem of identifying the maximal element of N reduces to the problem of sorting:

Let Φ be the identity algorithm and define Ψ to access and return the first element in the sorted ordering of N

Reduction Example

Let N be as before. The problem of identifying the maximal element of N reduces to the problem of sorting:

Let Φ be the identity algorithm and define Ψ to access and return the first element in the sorted ordering of N



Determining Decidability via Reduction

A common way to determine the decidability of a problem is via problem reduction.

Determining Decidability via Reduction

A common way to determine the decidability of a problem is via problem reduction.

For any problem P ,

If P reduces to a decidable problem, P is decidable.

If an undecidable problem reduces to P , P is undecidable.

Determining Decidability via Reduction

A common way to determine the decidability of a problem is via problem reduction.

For any problem P ,

If P reduces to a decidable problem, P is decidable.

If an undecidable problem reduces to P , P is undecidable.

We will show that the fractal geometry problem is undecidable via reduction from an undecidable problem.

An Undecidable Problem (notational prerequisites)

Let Σ be a finite alphabet (list of characters).

An Undecidable Problem (notational prerequisites)

Let Σ be a finite alphabet (list of characters).

Σ^* is the collection of all words (finite strings) made from Σ .

An Undecidable Problem (notational prerequisites)

Let Σ be a finite alphabet (list of characters).

Σ^* is the collection of all words (finite strings) made from Σ .

Σ^ω is the collection of all infinite strings made from Σ .

An Undecidable Problem (notational prerequisites)

Let Σ be a finite alphabet (list of characters).

Σ^* is the collection of all words (finite strings) made from Σ .

Σ^ω is the collection of all infinite strings made from Σ .

$\Sigma^\infty = \Sigma^* \cup \Sigma^\omega$ is the collection of all strings (finite or infinite) made from Σ

An Undecidable Problem (notational prerequisites)

Let Σ be a finite alphabet (list of characters).

Σ^* is the collection of all words (finite strings) made from Σ .

Σ^ω is the collection of all infinite strings made from Σ .

$\Sigma^\infty = \Sigma^* \cup \Sigma^\omega$ is the collection of all strings (finite or infinite) made from Σ

Example

Let $\Sigma = \{1, 2, 3, 4\}$. Then $1312423 \in \Sigma^*$, $111\dots \in \Sigma^\omega$ and both $1312423, 111\dots \in \Sigma^\infty$

An Undecidable Problem (notational prerequisites)

Let Σ be a finite alphabet (list of characters).

Σ^* is the collection of all words (finite strings) made from Σ .

Σ^ω is the collection of all infinite strings made from Σ .

$\Sigma^\infty = \Sigma^* \cup \Sigma^\omega$ is the collection of all strings (finite or infinite) made from Σ

Example

Let $\Sigma = \{1, 2, 3, 4\}$. Then $1312423 \in \Sigma^*$, $111\dots \in \Sigma^\omega$ and both $1312423, 111\dots \in \Sigma^\infty$

Note

For our purposes, we assume Σ is composed of the positive integers up to some k . For any other alphabet, we work with an enumeration of its characters rather than the characters themselves.

An Undecidable Problem

I-PCP (The Infinite Post Correspondence Problem)

Given lists A and B of k words each from Σ^* , say

$$A = a_1, a_2, \dots, a_k \quad \text{and} \quad B = b_1, b_2, \dots, b_k$$

does there exist a sequence of integers $\{i_\bullet\} = \{i_1, i_2, \dots\}$ with $1 \leq i_\bullet \leq k$ such that

$$a_{i_1} a_{i_2} \cdots = b_{i_1} b_{i_2} \cdots ?$$

The sequence $i_\bullet$ is called a *solution* of this instance of I-PCP.

An Undecidable Problem (example problem instance)

Example

Let an instance of I-PCP be given by

$\Sigma = \{1, 2, 3\}$, $A = \{131, 2, 12\}$ and $B = \{1, 312, 12\}$.

An Undecidable Problem (example problem instance)

Example

Let an instance of I-PCP be given by

$\Sigma = \{1, 2, 3\}$, $A = \{131, 2, 12\}$ and $B = \{1, 312, 12\}$.

$i_{\bullet} = \{1, 2, 3, 3, 3, \dots\}$ is a solution, as

An Undecidable Problem (example problem instance)

Example

Let an instance of I-PCP be given by

$\Sigma = \{1, 2, 3\}$, $A = \{131, 2, 12\}$ and $B = \{1, 312, 12\}$.

$i_{\bullet} = \{1, 2, 3, 3, 3, \dots\}$ is a solution, as

$$A_{i_{\bullet}} = 131212\overline{12} = B_{i_{\bullet}}.$$

Reducing I-PCP to the Fractal Geometry Problem

I-PCP reduces to the fractal geometry problem by interpreting the infinite strings of I-PCP as addresses in the code space of some IFS acting on $[0, 1]^2$.

Reducing I-PCP to the Fractal Geometry Problem

I-PCP reduces to the fractal geometry problem by interpreting the infinite strings of I-PCP as addresses in the code space of some IFS acting on $[0, 1]^2$.

To identify this IFS, we need a 'good' map from infinite strings to real numbers.

Reducing I-PCP to the Fractal Geometry Problem

I-PCP reduces to the fractal geometry problem by interpreting the infinite strings of I-PCP as addresses in the code space of some IFS acting on $[0, 1]^2$.

To identify this IFS, we need a 'good' map from infinite strings to real numbers.

Note

This map must be good in the sense that it is well defined (does not map any string to more than 1 real number) and injects from Σ^ω (does not map multiple infinite strings to the same real number).

The Dube Interpretation Function [1]

Let $\Sigma = \{1, 2, 3, \dots, k\}$ for some $k \in \mathbb{N}$ and let $I : \Sigma^\infty \rightarrow [0, 1]$ denote the Dube interpretation function.

The Dube Interpretation Function [1]

Let $\Sigma = \{1, 2, 3, \dots, k\}$ for some $k \in \mathbb{N}$ and let $I : \Sigma^\infty \rightarrow [0, 1]$ denote the Dube interpretation function.

For the empty string ϵ , let $I(\epsilon) = 0$

The Dube Interpretation Function [1]

Let $\Sigma = \{1, 2, 3, \dots, k\}$ for some $k \in \mathbb{N}$ and let $I : \Sigma^\infty \rightarrow [0, 1]$ denote the Dube interpretation function.

For the empty string ϵ , let $I(\epsilon) = 0$

A non-empty string $w \in \Sigma^*$ (note that 0 is not in $\Sigma = \{1, \dots, k\}$) is interpreted as a rational number by putting the radix point to the left and interpreting the string in base $k + 1$ notation. We can similarly interpret infinite-strings as real numbers.

The Dube Interpretation Function [1]

Let $\Sigma = \{1, 2, 3, \dots, k\}$ for some $k \in \mathbb{N}$ and let $I : \Sigma^\infty \rightarrow [0, 1]$ denote the Dube interpretation function.

For the empty string ϵ , let $I(\epsilon) = 0$

A non-empty string $w \in \Sigma^*$ (note that 0 is not in $\Sigma = \{1, \dots, k\}$) is interpreted as a rational number by putting the radix point to the left and interpreting the string in **base $k + 1$** notation. We can similarly interpret infinite-strings as real numbers.

The Dube Interpretation Function [1]

Let $\Sigma = \{1, 2, 3, \dots, k\}$ for some $k \in \mathbb{N}$ and let $I : \Sigma^\infty \rightarrow [0, 1]$ denote the Dube interpretation function.

For the empty string ϵ , let $I(\epsilon) = 0$

A non-empty string $w \in \Sigma^*$ (note that 0 is not in $\Sigma = \{1, \dots, k\}$) is interpreted as a rational number by putting the radix point to the left and interpreting the string in base $k + 1$ notation. We can similarly interpret infinite-strings as real numbers.

Example

Let $\Sigma = \{1, 2, 3, 4\}$. A few strings' interpretations follow:

- ▶ $I(\epsilon) = 0$
- ▶ $I(1234) = 0.1234$
- ▶ $I(111\dots) = 0.111\dots$

I-PCP's associated IFS on $[0, 1]^2$

With the Dube interpretation function, we are able to identify the IFS relevant to our reduction from I-PCP.

I-PCP's associated IFS on $[0, 1]^2$

With the Dube interpretation function, we are able to identify the IFS relevant to our reduction from I-PCP.

Fix an instance of I-PCP. That is, fix Σ , $k \in \mathbb{N}$, $A = \{a_1, a_2, \dots, a_k\}$ and $B = \{b_1, b_2, \dots, b_k\}$ such that $a_\bullet, b_\bullet \in \Sigma^*$.

I-PCP's associated IFS on $[0, 1]^2$

With the Dube interpretation function, we are able to identify the IFS relevant to our reduction from I-PCP.

Fix an instance of I-PCP. That is, fix Σ , $k \in \mathbb{N}$, $A = \{a_1, a_2, \dots, a_k\}$ and $B = \{b_1, b_2, \dots, b_k\}$ such that $a_\bullet, b_\bullet \in \Sigma^*$.

Let $\mathcal{W} \curvearrowright [0, 1]^2$ by

$$\mathcal{W} = \left\{ f_j \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} (k+1)^{-|a_j|}x + l(a_j) \\ (k+1)^{-|b_j|}y + l(b_j) \end{pmatrix} : 1 \leq j \leq k \right\}$$

Reducing I-PCP to the Fractal Geometry Problem

Fix an instance of I-PCP. That is, fix Σ , $k \in \mathbb{N}$,
 $A = \{a_1, a_2, \dots, a_k\}$ and $B = \{b_1, b_2, \dots, b_k\}$ such that
 $a_\bullet, b_\bullet \in \Sigma^*$.

There exists a solution to this instance of I-PCP if and only if there
exists an $\{i_\bullet\}$ such that $a_{i_1} a_{i_2} \dots = b_{i_1} b_{i_2} \dots$ if and only if
 $\{(a_{i_1} a_{i_2} \dots, b_{i_1} b_{i_2} \dots) : 1 \leq i_\bullet \leq k\}$ has a diagonal point.

It can be shown that $\{(a_{i_1} a_{i_2} \dots, b_{i_1} b_{i_2} \dots) : 1 \leq i_\bullet \leq k\}$ bijects
component-wise to $A_{\mathcal{W}}$ through I and thus
 $\{(a_{i_1} a_{i_2} \dots, b_{i_1} b_{i_2} \dots) : 1 \leq i_\bullet \leq k\}$ has a diagonal point if and
only if $A_{\mathcal{W}}$ has a diagonal point.

Finally, $A_{\mathcal{W}} \subset [0, 1]^2$ has a diagonal point if and only if it
intersects the line segment from $\langle 0, 0 \rangle$ to $\langle 1, 1 \rangle$.

Reducing I-PCP to the Fractal Geometry Problem

Fix an instance of I-PCP. That is, fix Σ , $k \in \mathbb{N}$, $A = \{a_1, a_2, \dots, a_k\}$ and $B = \{b_1, b_2, \dots, b_k\}$ such that $a_\bullet, b_\bullet \in \Sigma^*$.

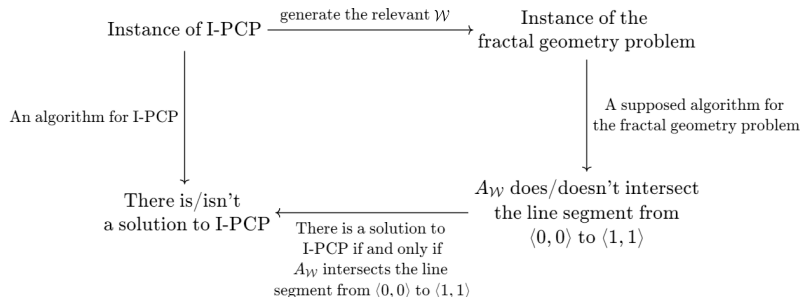
There exists a solution to this instance of I-PCP if and only if there exists an $\{i_\bullet\}$ such that $a_{i_1} a_{i_2} \dots = b_{i_1} b_{i_2} \dots$ if and only if $\{(a_{i_1} a_{i_2} \dots, b_{i_1} b_{i_2} \dots) : 1 \leq i_\bullet \leq k\}$ has a diagonal point.

It can be shown that $\{(a_{i_1} a_{i_2} \dots, b_{i_1} b_{i_2} \dots) : 1 \leq i_\bullet \leq k\}$ bijects component-wise to $A_{\mathcal{W}}$ through I and thus $\{(a_{i_1} a_{i_2} \dots, b_{i_1} b_{i_2} \dots) : 1 \leq i_\bullet \leq k\}$ has a diagonal point if and only if $A_{\mathcal{W}}$ has a diagonal point.

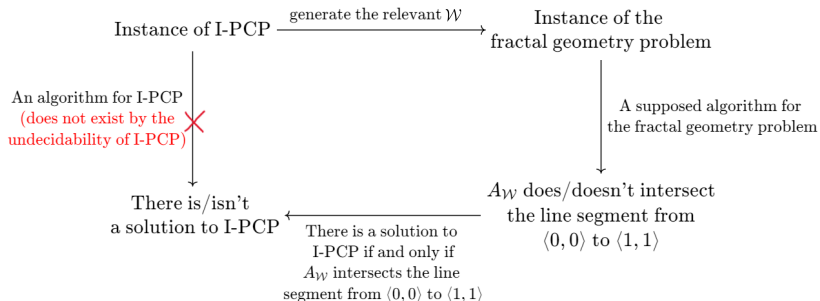
Finally, $A_{\mathcal{W}} \subset [0, 1]^2$ has a diagonal point if and only if it intersects the line segment from $\langle 0, 0 \rangle$ to $\langle 1, 1 \rangle$.

In particular, there exists a solution to this instance of I-PCP if and only if $A_{\mathcal{W}} \subset [0, 1]^2$ intersects the line segment from $\langle 0, 0 \rangle$ to $\langle 1, 1 \rangle$

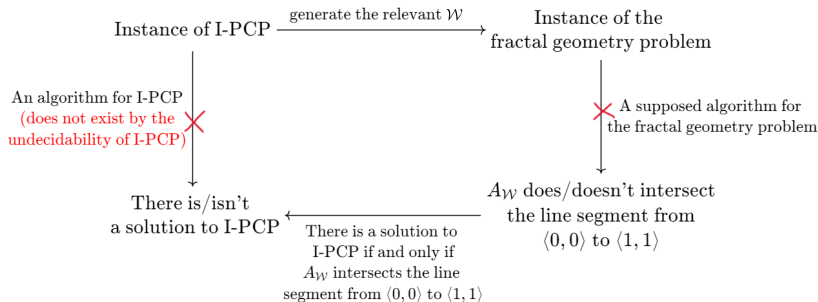
Reducing I-PCP to the Fractal Geometry Problem



Reducing I-PCP to the Fractal Geometry Problem



Reducing I-PCP to the Fractal Geometry Problem



The Undecidability of the Fractal Geometry Problem

Since I-PCP reduces to the fractal geometry problem and I-PCP is undecidable, the fractal geometry problem is undecidable.

The Undecidability of the Fractal Geometry Problem

Since I-PCP reduces to the fractal geometry problem and I-PCP is undecidable, the fractal geometry problem is undecidable.

That is, there does not exist an algorithm which (in finite time) determines whether or not the attractor of any given iterated function system intersects the line segment between $\langle 0, 0 \rangle$ and $\langle 1, 1 \rangle$.

Each $f_j \in \mathcal{W}$ left concatenates (in the interpretation space) its argument (an interpreted string pair) by $I \begin{pmatrix} a_j \\ b_j \end{pmatrix}$:

Suppose $\Sigma = \{1, 2, 3, 4\}$ and $(a_j, b_j) = (1, 22)$ so that $I(a_j, b_j) = (.1, .22)$. Notice

$$\begin{aligned} f_j \left(I \begin{pmatrix} 333 \\ 4444 \end{pmatrix} \right) &= \begin{pmatrix} (5)^{-|1|}(.333) + .1 \\ (5)^{-|2|}(.4444) + .22 \end{pmatrix} = \begin{pmatrix} .1333 \\ .224444 \end{pmatrix} \\ &= I \begin{pmatrix} 1333 \\ 224444 \end{pmatrix} \\ &= I \begin{pmatrix} a_j 333 \\ b_j 4444 \end{pmatrix} \end{aligned}$$

$$\begin{array}{ccc}
 \begin{array}{c} A = \{a_1, a_2, \dots, a_k\} \\ \text{and} \\ B = \{b_1, b_2, \dots, b_k\} \end{array} & \xrightarrow{I} & \begin{array}{c} I(A) = \{I(a_1), I(a_2), \dots, I(a_k)\} \\ \text{and} \\ I(B) = \{I(b_1), I(b_2), \dots, I(b_k)\} \end{array} \\
 \downarrow \text{Infinite Concatenation IFS} & & \downarrow \mathcal{W} \\
 \{(a_{i_1} a_{i_2} \dots, b_{i_1} b_{i_2} \dots) : 1 \leq i_{\bullet} \leq k\} & \xleftarrow{I^{-1}} & A_{\mathcal{W}}
 \end{array}$$